Deep Learning

Cross-Entropy Loss

Classification

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3. In other words, we don't prefer MSE loss for learning

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4. Loss function should compare **p** and **q**

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3. Hence, the information can be calculated as $I(x) = -log_2(P(x))$
4. This is also the number of bits required to encode x

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2. Expected amount of information in an event drawn from that distribution $H(X) = \mathbb{E}_{x \sim p}[I(x)]$

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4. Skewed distribution has less entropy, uniform/balanced distribution has more entropy

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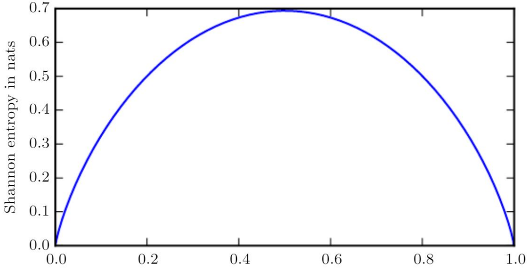

Entropy for a binary random variable

Figure credits Goodfellow et al. 2016

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4. Cross-entropy between a distribution and itself $(H(p,q))$ gives the entropy of the distribution $H(p)$

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2. $H(p,q) = H(p) + KL(p||q)$ where $KL(p||q) = \sum p_i \cdot log\left(\frac{p_i}{q_i}\right)$

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3. Target distribution (or, groundtruth) is one-hot encoding p, and model predicts a distribution q

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   - make them probabilities (i.e. sum to 1)

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2. $(\alpha_1, \alpha_2, \ldots, \alpha_C) \to (q_1, q_2, \ldots, q_C)$

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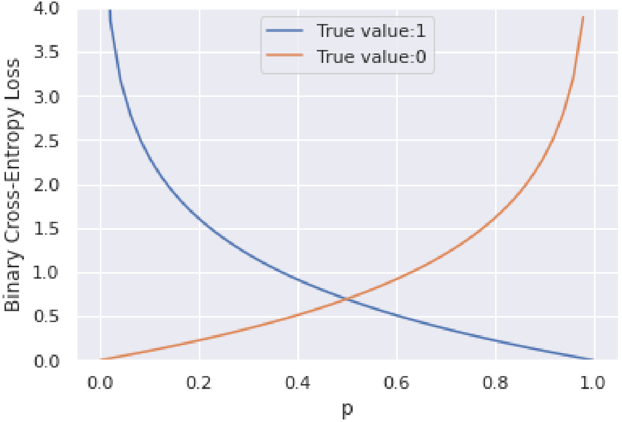
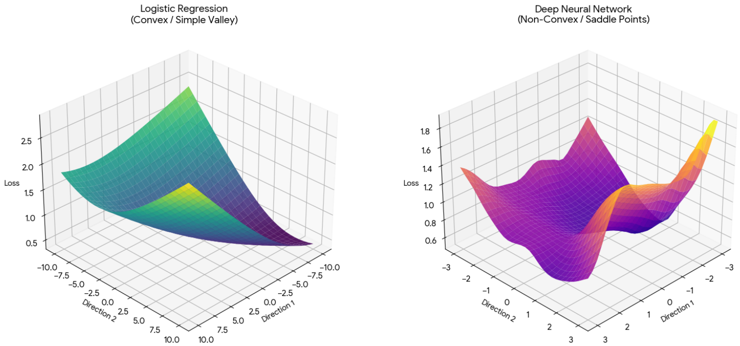
   - large if the model assigns smaller probability for the groundtruth class $(q_c \approx 0)$

BCE



True value:1
True value:0
Binary Cross-Entropy Loss
p
0.0
0.2
0.4
0.6
0.8
1.0
0.5
1.5
2.0
2.5
3.0
3.5
4.0

Loss surface



Loss surface : Logistic regression vs Neural networks

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https://colab.research.google.com/drive/1TnUFFUhXCNfw1bUA2s9k8LRCfrIzDpXL?usp=sharing

Li et al., 2018 Visualizing the Loss Landscape of Neural Nets

Logistic regression vs Neural Networks

$$L(w) = -[y \log(\hat{y}) + (1-y)\log(1-\hat{y})]$$

$$\hat{y} = \sigma(w^T x)$$

$$\nabla L(w) = (\hat{y} - y)x$$

$$\nabla^2 L(w) = \frac{\partial}{\partial w}[(\hat{y} - y)x]$$

$$\nabla^2 L(w) = \hat{y}(1-\hat{y})xx^T$$

Consider a simple network with one hidden layer containing 2 neurons, A and B
- weights W_{opt} that reaches a global minimum
- obtain W'_{opt} by swapping the weights of neuron A and neuron B (and their outgoing weights)

$$L(W_{opt}) = L(W'_{opt})$$

$$L(\frac{W_{opt} + W'_{opt}}{2}) > L(W_{opt})$$